



Project-Based Learning

Phase-I Evaluation

Intelligent Host Security System: HIDS

Team ID: DSCPP-III-2026-T187

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun

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Team & Mentor Details

Team Information

Team ID: DSCPP-III-2026-T187

Semester: 3rd

Section: DS1/DS2/ML2/J

Domain: DSCPP

Team Members

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Mentor

Mr. Ramesh Singh Rawat

Interactions completed: 3

Problem Statement & Motivation for HIDS

Problem Statement

Host systems continuously generate processes, file activities, user activities and system events. Suspicious behaviour may indicate malware, unauthorized access or other security threats. The proposed Host-Based Intrusion Detection System will monitor selected host activities and identify suspicious patterns using predefined detection rules.

Motivation

- ▶ Detect suspicious activities occurring on a host system
- ▶ Monitor important system activities and events
- ▶ Generate alerts when predefined suspicious patterns occur
- ▶ Maintain logs for analysis and reporting

Objectives

1. Develop a working Host-Based Intrusion Detection System in C++.
2. Monitor selected processes, files and system events.
3. Detect suspicious activities using predefined rules.
4. Assign threat scores and priorities to detected events.
5. Generate alerts and maintain event logs.
6. Demonstrate the use of OOP and DSA concepts in a real-world cybersecurity application.

Proposed Solution for HIDS

Monitoring

- ▶ Process monitoring
- ▶ File activity monitoring
- ▶ System event monitoring

Detection

- ▶ Rule-based detection
- ▶ Suspicious event identification
- ▶ Threat score calculation

Response

- ▶ Priority-based alerts
- ▶ Event logging
- ▶ Suspicious activity reports

Implementation

- ▶ C++
- ▶ Object-Oriented Programming
- ▶ Data Structures & Algorithms
- ▶ Windows APIs

Integration of PBL Course Concepts

Object-Oriented Programming

- ▶ Classes and objects
- ▶ Encapsulation
- ▶ Abstraction
- ▶ Inheritance
- ▶ Polymorphism

Data Structures & Algorithms

- ▶ Queues and Priority Queue
- ▶ Hash Table
- ▶ Graph
- ▶ Searching and sorting
- ▶ Rule matching

Application

OOP will organize monitoring and detection modules, while DSA will support event storage, rule matching, threat prioritization and efficient searching.

System Architecture / Workflow



Major Components

- ▶ Monitoring Module (Process, File, Registry, Login, Network)
- ▶ Event Collection and Normalization
- ▶ Detection & Analysis Engine (Rule-based + DSA)
- ▶ Alert Generation
- ▶ Logging & Storage (File Handling)
- ▶ User Interface (Console-Based)

Data / Control Flow

- ▶ Host activities are monitored in real-time and events are collected.
- ▶ Events are normalized and analyzed using predefined rules and DSA algorithms.
- ▶ If a suspicious pattern is detected, an alert is generated and assigned a severity level.
- ▶ All events and alerts are logged and stored in files.
- ▶ Detected alerts and logs are displayed to the user through a console-based interface.

Technology Stack & Methodology

Technology Stack

- ▶ C++
- ▶ Object-Oriented Programming
- ▶ Data Structures & Algorithms
- ▶ Windows APIs
- ▶ C++ STL
- ▶ CMake
- ▶ Git & GitHub

Methodology

1. Requirement analysis
2. System architecture design
3. Module development
4. Rule-based detection
5. Testing with sample events
6. Integration and final implementation

Initial Progress & Team Contribution

Progress Achieved

- ▶ Requirements for HIDS identified
- ▶ System architecture and workflow prepared
- ▶ Monitoring modules and detection rules planned
- ▶ DSA structures mapped to project modules
- ▶ GitHub repository initialised for version control

Team Approach

- ▶ Module-wise responsibility
- ▶ Regular discussions and code reviews
- ▶ Parallel development and integration
- ▶ Shared documentation and version control

Individual Contribution

Member	Role	Completed Work / Contribution
Tanish Sood	Team Lead / DSA & Detection	HIDS architecture and workflow design, DSA mapping, detection rules and detection-engine planning
Shambhavi sri-vastava	OOP / Monitoring Module	OOP class/module design, monitoring-module planning, process/file/system activity identification and Windows API requirements
Jitender singh	OOP / Event & Logging	Event collection and normalization design, SecurityEvent class structure, file-logging and storage approach
Chirag Varshney	Alert / UI & Testing	Alert-generation workflow, console UI planning, sample-event testing and validation approach

Roadmap, Expected Outcomes & References

Roadmap

1. Phase-I: Research & System Design
2. Phase-II: Module Development
3. Phase-III: Testing & Final Implementation

Expected Outcomes

- ▶ Working Host-Based Intrusion Detection System with real-time monitoring of processes, files and system events
- ▶ Detection of suspicious activities using predefined rules and threat scoring
- ▶ Priority-based alert generation for detected threats
- ▶ Event logging and generation of suspicious activity reports

References

1. Microsoft Windows API Documentation
2. C++ STL Documentation
3. Windows Security and Event Logging Documentation
4. CMake Documentation
5. Git & GitHub Documentation

Thank You

Questions & Suggestions